Formalizing Fefferman's Official Periodic Navier–Stokes Problem (B) in Lean 4: Global Linear Strain Exclusion on $\mathbb{T}^3$, Biot–Savart Direction Depletion Identities, and Conditional Reductions via Beale–Kato–Majda

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October 27, 2026; revised November 11, 2026 (v1.5.0 erratum)

Abstract

We present a self-contained Lean 4 formalization (`leanprover/lean4:v4.30.0-rc1`, Mathlib 4, 236 theorem/lemma declarations in source across 7 modules with zero `sorry` and zero non-standard axioms outside `[propext, Classical.choice, Quot.sound]`) of Charles L. Fefferman's official periodic 3D incompressible Navier–Stokes problem statement (*Existence and Smoothness of the Navier–Stokes Equation*, Clay Mathematics Institute, 2006, Statement (B) on $\mathbb{R}^3/\mathbb{Z}^3$) together with unconditional geometric/kinematic identities, spatial-maximum Bochner–polar reduction, and explicit **conditional reductions** to Beale–Kato–Majda continuation. In `FeffermanClayNS/FinalProductTheorem.lean`, five proved components (Q_1–Q_3 unconditional; Q_4 and Q_5 implications from stated hypotheses) are packaged into the single 5-conjunct master product theorem `fefferman_periodic_ns_exact_reduction_product_` $(Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu))$.

Erratum (v1.5.0, November 11, 2026). In v1.0–v1.4 the open hypothesis of Q_5, namely `PeriodicDirectionDepletionBound` ν, required a two-sided derivative (`DifferentiableAt`) of $M(t) = \|\omega(t)\|_\infty$ at every $t \in [0,T]$, including $t = 0$. But the solution predicate `IsPeriodicSmoothSolutionOn` does not constrain $u(t)$ for $t < 0$, so the path `jumpPath`, equal to the zero solution on $[0,\infty)$ and to a linear shear with nonzero constant vorticity for $t < 0$, is a solution on $[0,1]$ whose M jumps at 0. The kernel-checked theorem `old_periodicDirectionDepletionBound_unsatisfiable` proves $\neg$ `PeriodicDirectionDepletionBoundOriginal` ν for every ν (the old form is kept under that name). Hence Q_5 and `clay_B_exact_of_torus_direction_depletion` were vacuously true in v1.0–v1.4. From v1.5.0 the hypothesis requires continuity of M on $[0,T]$ and a right derivative d at each $s \in [0,T)$ with $d \le C_{\mathrm{CZ}} M(s)^2 - \kappa'' M(s)^3$; the ODE ceiling is `comparison_ode_ceiling_right`. The regression `jumpPath_oneSided_depletion_clause` shows the counterexample path satisfies the corrected trajectory clause; it does not show the corrected hypothesis holds in general, which remains open. All statements below refer to the corrected definition. Nothing here resolves the Clay problem.

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1 Introduction and Single-Sentence Final Product Form $(Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu))$

In his official Clay Mathematics Institute problem description [1], Charles L. Fefferman formulates the periodic existence and smoothness problem (Statement (B)) for the three-dimensional incompressible Navier–Stokes equations on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. For the unmodified Navier–Stokes PDE ($\kappa = 0$ in the dynamics), Lean's kernel cannot derive $\langle \omega, (\omega\cdot\nabla)u\rangle + \nu\langle \omega, \Delta\omega\rangle \le C_{\mathrm{CZ}} M^3 - \kappa'' M^4$ at a spatial maximum without either taking the high-set direction-curvature bound $|\nabla\hat{\xi}(x_{\max})|^2 \ge c_{\mathrm{dir}} M$ as an explicit hypothesis or proving that open PDE estimate from scratch. Accordingly, this repository states **only the theorems it proves explicitly and unconditionally**, packaged in `FeffermanClayNS/FinalProductTheorem.lean` into a single 5-conjunct product theorem.

1.1 Single-Sentence English Prose Statement

Theorem 1.1 (Master Product Theorem: `fefferman_periodic_ns_exact_reduction_product_theorem`). *For every viscosity $\nu \in \mathbb{R}$ on the 3-torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, **($Q_1$)** spatial $\mathbb{Z}^3$-periodicity is equivalent to invariance under the three unit coordinate shifts $f(x + e_j) = f(x)$, and Fefferman's energy-bounded periodic statement implies his exact periodic statement (*$\mathrm{Clay_B_exact_with_energy}(\nu) \implies \mathrm{Clay_B_exact}(\nu)$*); **($Q_2$)** every $\mathbb{Z}^3$-periodic linear map $L : \mathbb{R}^3 \to \mathbb{R}^3$ vanishes identically ($L = 0$), excluding any non-zero global linear Burgers background strain* $\mathrm{diag}(-\alpha/2, -\alpha/2, \alpha)x$ *($\alpha \neq 0$) on $\mathbb{T}^3$; **(Q_3)** the Biot–Savart vortex-stretching determinant obeys* $\det(\omega(y), z, \omega(x)) = \det(\omega(y) - \omega(x), z, \omega(x))$ *and vanishes under collinear alignment $\omega(y) = c\,\omega(x)$, while under parabolic rescaling $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$ both cubic stretching $C_{\mathrm{CZ}}|\omega_\lambda|^3$ and viscous direction-curvature dissipation $\nu|\omega_\lambda|^2|\nabla\hat{\xi}_\lambda|^2$ scale homogeneously as λ^6; **(Q_4)** at a spatial maximum $x_{\max}$ where $\langle \omega, (\omega\cdot\nabla)u\rangle \le C_{\mathrm{CZ}} M^3$, $\nu\langle \omega, \Delta\omega\rangle \le -\nu M^2 |\nabla\hat{\xi}|^2$, and high-set direction-curvature coercivity $|\nabla\hat{\xi}(x_{\max})|^2 \ge c_{\mathrm{dir}} M$ holds, the maximum vorticity rate satisfies $M'(t) \le C_{\mathrm{CZ}} M(t)^2 - (\nu c_{\mathrm{dir}}) M(t)^3$, which forces the uniform Lyapunov ceiling $M(t) \le \max(M(0), C_{\mathrm{CZ}}/(\nu c_{\mathrm{dir}}))$; and **($Q_5$)** whenever a periodic flow class satisfies the high-set direction-curvature depletion hypothesis* $\mathrm{PeriodicDirectionDepletionBound}(\nu)$ *(one-sided form, v1.5.0) and classical periodic Beale–Kato–Majda continuation* $\mathrm{PeriodicBKMContinuation}(\nu)$*, Fefferman's periodic statement* $\mathrm{Clay_B_exact}(\nu)$ *holds.*

1.2 Exact Mathematical Product Form ($Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu)$)

$$\mathcal{Q}_{\mathrm{ClayB}}(\nu) \;:\Longleftrightarrow\; \underbrace{\Big((\forall f : \mathbb{R}^3 \to \mathbb{R}^3,\ \mathrm{IsZ3Periodic}(f) \iff \forall x,\ j \in \{1,2,3\},\ f(x+e_j) = f(x)) \ \wedge\ (\mathrm{Clay_B_exa}}_{Q_1(\nu)\text{: Exact Fefferman Periodic Unit-Shift Equivalence \& Energy I}}$$

$$\wedge\ \underbrace{\Big((\forall L : \mathbb{R}^3 \to_{\mathbb{R}} \mathbb{R}^3,\ \mathrm{IsZ3Periodic}(L) \implies L = 0) \ \wedge\ (\forall \alpha \neq 0,\ \neg\mathrm{IsZ3Periodic}(\mathrm{diag}(-\tfrac{\alpha}{2}, -\tfrac{\alpha}{2}}_{Q_2\text{: Torus Exclusion of Global Linear Burgers Strain}}$$

$$\wedge\ \underbrace{\Big(\det(\omega_y, z, \omega_x) = \det(\omega_y - \omega_x, z, \omega_x) \ \wedge\ \det(c\omega_x, z, \omega_x) = 0 \ \wedge\ C_{\mathrm{CZ}}(\lambda^2|\omega|)^3 - \nu(\lambda^2|\omega|)^2(\lambda|}_{Q_3\text{: Biot–Savart Alignment Depletion \& } \lambda^6 \text{ Parabolic Scaling Homogene}}$$

$$\wedge\ \underbrace{\Big((\mathrm{Stretch} \leq C_{\mathrm{CZ}} M^3 \wedge \mathrm{Diff} \leq -\nu M^2 |\nabla\hat{\xi}|^2 \wedge |\nabla\hat{\xi}|^2 \geq c_{\mathrm{dir}} M \implies M' \leq C_{\mathrm{CZ}} M^2 - \nu c_{\mathrm{dir}} M^3}_{Q_4\text{: Spatial-Maximum Bochner–Polar Reduction \& C}}$$

$$\wedge\ \underbrace{\Big((\mathrm{PeriodicDirectionDepletionBound}(\nu) \ \wedge\ \mathrm{PeriodicBKMContinuation}(\nu)) \implies \mathrm{Clay_B_}}_{Q_5(\nu)\text{: Exact Conditional Implication to Fefferman's Clay_B_exact}(\nu)} \tag{1}$$

2 Verbatim Lean 4 Kernel Audit and Cryptographic Attestation

A comment-aware count of `theorem`/`lemma` declaration headers in the Lean sources (the `python3` command in the repository README) gives 236 declarations across 7 modules. `AxiomAudit.lean` audits 243 names (those 236 theorem/lemma declarations together with structure fields and definitions). The titan audit reports more because it counts compiled/generated constants. The master product theorem depends only on the standard axioms `[propext, Classical.choice, Quot.sound]`.

Listing 1: Source `theorem`/`lemma` declaration counts in `fefferman-clay-navier-stokes` (not a titan-audit or AxiomAudit total).

```
FeffermanClayNS.Defs: theorems=131
FeffermanClayNS.FeffermanStatements: theorems=10
FeffermanClayNS.FinalProductTheorem: theorems=21
FeffermanClayNS.HighSetDepletionAndBKM: theorems=38
FeffermanClayNS.MainTheorem: theorems=1
FeffermanClayNS.Stage6Calculus: theorems=16
FeffermanClayNS.TorusStrainExclusion: theorems=19
TOTAL theorems=236
'FeffermanClayNS.fefferman_periodic_ns_exact_reduction_product_theorem' depends on axioms: [
    propext, Classical.choice, Quot.sound]
```

A Verification certificate

The formal development was machine-checked as follows.

- Repository:
  https://github.com/BenFrohman/fefferman-clay-navier-stokes
- Verified commit:
  `8c40f83d7d8b52b2141506063952e2ebb3ed425a`
- Lean toolchain: `leanprover/lean4:v4.30.0-rc1`

Table 1: Summary of Unconditionally Proved Theorems vs. Open Analytic Hypotheses in `FeffermanClayNS` (236 theorem/lemma declarations, 0 `sorry`).

Declaration / Mathematical Item	Status	Mathematical Meaning & Scope
`fefferman_periodic_ns_exact_reduction_product_theorem` (`FinalProductTheorem.lean`)	**PROVED** (Unconditional, 21 theorem/lemma declarations)	Master 5-conjunct product theorem ($Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu)$) and spatial-maximum Bochner–polar reduction.
`Clay_B_exact` ν, `isZ3Periodic_iff_unit_shifts`, `clay_B_exact_of_with_energy`	**PROVED** (Unconditional)	Exact encoding of Fefferman's Statement (B), unit-shift periodicity equivalence, and logical implication from the energy-bounded variant.
`z3_periodic_linear_eq_zero`, `burgersLinearStrain_not_z3_periodic`	**PROVED** (Unconditional)	Any $\mathbb{Z}^3$-periodic linear map on $\mathbb{R}^3$ vanishes identically; the global linear Burgers strain $\mathrm{diag}(-\alpha/2, -\alpha/2, \alpha)x$ ($\alpha \neq 0$) is not $\mathbb{Z}^3$-periodic.
`biot_savart_det_aligned_eq_zero`, `aligned_vorticity_stretching_integrand_zero`, `biot_savart_det_difference_identity`	**PROVED** (Unconditional)	Exact pointwise vanishing of $\det(\omega(y), z, \omega(x))$ when $\omega(y) = c\,\omega(x)$, reducing the stretching kernel to direction increments $\omega(y) - \omega(x)$.
`first_remaining_gap_bochner_polar_route_of_maximum_at_max`, `scale_invariant_cubic_rate_of_direction_coercivity`, `comparison_ode_ceiling`, `comparison_ode_ceiling_right`	**PROVED** (Unconditional)	Spatial-maximum Bochner–polar reduction from $\lvert\nabla\hat{\xi}\rvert^2 \geq c_{\mathrm{dir}}\lvert\omega\rvert$ to $M' \leq C_{\mathrm{CZ}}M^2 - \kappa'' M^3$ and ODE comparison ceiling $M(t) \leq \max(M(0), C_{\mathrm{CZ}}/\kappa'')$.
`clay_B_exact_of_periodic_tether_bound`, `clay_B_exact_of_torus_direction_depletion`	**PROVED CONDITIONAL REDUCTIONS**	Conditional theorems proving that *if* the open direction-depletion/tether hypothesis and BKM continuation hold, then `Clay_B_exact` ν follows.
`periodicTstar_eq_top_of_bkm`, `periodicGlobalOfTstarTop`, `clay_B_exact_of_periodic_bkm`	**PROVED CONDITIONAL REDUCTION**	Smooth solutions on $[0,T]$ carry no $L^2(\mathbb{R}^3)$ clause (it fails for every nonzero periodic field; the earlier definitions, kept with suffix `Original`, made `PeriodicLocalExistence` and `PeriodicMaximalSolution` false, as proved by `not_periodicLocalExistenceOriginal`, `not_periodicMaximalSolutionOriginal`), and maximality is required only for $0 < T$. With $T^* = \sup\{T :$ a periodic smooth solution exists on $[0,T]\}$ in $\overline{\mathbb{R}}$: `PeriodicLocalExistence`, `PeriodicMaximalSolution`, `PeriodicBKMCriterion` (classical, not formalized; the global step `periodicGlobalOfTstarTop` is proved) and `PeriodicBKMIntegrability` (OPEN: $\int_0^T \lVert\omega\rVert_\infty < \infty$ for every T along the maximal solution) $\Rightarrow T^* = \infty \Rightarrow$ `Clay_B_exact`.
`periodicFiniteTimeBound_vacuous`, `periodicFiniteTimeVorticityBound_of_smooth`, `periodicBKMContinuation_of_finiteTime_scaffold`, `clay_B_exact_of_periodic_finite_time_bound_scaffold`	**PROVED (SCAFFOLDING)**	The vorticity-bound Props quantify only over solutions already existing on $[0,T]$ and hold with $C = 0$ where none exists, so in these reductions the continuation hypothesis carries all of Statement (B).
`old_periodicDirectionDepletionBound_unsatisfiable`, `jumpPath_oneSided_depletion_clause`	**PROVED (v1.5.0 erratum)**	The v1.0–v1.4 two-sided form is false for every ν; the counterexample path satisfies the corrected one-sided clause.

- Mathlib: tag `v4.30.0-rc1`, commit (from `lake-manifest.json`)
  `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  https://github.com/BenFrohman/fefferman-clay-navier-stokes/actions/runs/43090450926
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  https://github.com/BenFrohman/fefferman-clay-navier-stokes/archive/8c40f83d7d8b52b2141506063952e2ebb3ed425a.tar.gz
  `6519b0aef51e3d31791f0ea449445e224f2eb9893f1d2997f6887ebcc191f1e2`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the commit archived on Zenodo for v1.5.0.

The development contains no `sorry` and no user-declared axioms. `AxiomAudit.lean` audits the 243 names listed below, of which 236 are theorem or lemma declarations in source; the remaining names are structure fields and definitions. Every one of these 243 names depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	#print axioms
FeffermanClayNS.AttainedSpatialMaximum.attained	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.AttainedSpatialMaximum.upper_bound	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.AttainedSpatialMaximum.vorticity_sup_norm_eq	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.aligned_vorticity_stretching_integrand_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.bddAbove_range_norm_of_attained_max	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.beltrami_amplitude_pos_and_bounded	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.beltrami_lamb_vector_cross3_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.biot_savart_det_aligned_eq_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.biot_savart_det_difference_identity	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.bochner_laplacian_3d_identity	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.burgersLinearStrain_coord	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.burgersLinearStrain_div_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.burgersLinearStrain_not_z3_periodic	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.burgers_background_strain_not_z3_periodic	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_B_exact_of_periodic_bkm	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_B_exact_of_periodic_finite_time_bound_scaffold	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_B_exact_of_periodic_tether_bound	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_B_exact_of_torus_direction_depletion	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
FeffermanClayNS.clay_B_exact_of_with_energy	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_reduces_to_direction_curvature_bound	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.comparison_ode_ceiling	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.comparison_ode_ceiling_right	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.concrete_bochner_polar_viscous_damping_at_max	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constE0_initialData	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_convective_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_isPeriodicSmoothInitialData	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_isPeriodicSmoothSolution	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_isZ3Periodic	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_laplacian_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_ne_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constantin_fefferman_near_far_cutoff_bridge	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.cross3_self	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.cubic_comparison_field_eq	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.cubic_comparison_field_neg_of_gt	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.curl_shearField	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.curl_shearField_ne_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.det3_self_right	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.directionalCoord_clm	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.directionalCoord_shearField	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.fefferman_periodic_ns_exact_reduction_product_theorem	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.first_remaining_gap_bochner_polar_damping_at_max	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.isPeriodicSmoothSolution_of_forall_horizon	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.isZ3Periodic_iff_unit_shifts	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.jumpPath_isPeriodicSmoothSolutionOn	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.jumpPath_of_nonneg	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.jumpPath_oneSided_depletion_clause	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.jumpPath_sup_norm	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.library_huniq_unsatisfiable_at_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.linear_strain_not_z3_periodic	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
FeffermanClayNS.navier_stokes_scaling_homogeneity_deg6	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.navier_stokes_vorticity_dt_scaling_exponent	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.near_field_stretching_absorbed_by_viscous_direction_damping	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.nonzero_periodic_smooth_solution_exists	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.not_differentiableAt_jump	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.not_isPhysicalNSSolution_of_constE0	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.not_periodicLocalExistenceOriginal	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.not_periodicMaximalSolutionOriginal	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.old_periodicDirectionDepletionBound_unsatisfiable	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicBKMContinuation_of_finiteTime_scaffold	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicExistenceTimesOriginal_constE0	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicFiniteTimeBound_vacuous	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicFiniteTimeVorticityBound_of_uniform	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicGlobalOfTstarTop	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicTetherClosingBound_of_directionDepletion	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicTstar_eq_top_of_bkm	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.polar_pythagoras_3d_sum	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.polar_pythagoras_directional_deriv	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.repaired_hInt	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.repaired_huniq_at_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.repaired_huniq_shift_at_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.scale_invariant_cubic_rate_of_direction_coercivity	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.stepField_not_physical	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.t3_eq_sum_stdBasisVec	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.unitCube_nonempty	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.unmodified_max_rate_of_cubic_source	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.unnormalized_quartic_scaling_mismatch	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.viscous_direction_dissipation_bridge	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.vorticity_sup_norm_eq_of_isSpatialMax	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.vorticity_sup_norm_shearField_pos	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.vorticity_sup_norm_zero_field	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
FeffermanClayNS.z3_periodic_continuousLinear_eq_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.z3_periodic_coord_diff_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.z3_periodic_deriv_integral_unit_interval_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.z3_periodic_linear_eq_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.z3_periodic_unit_shift	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.zero_isPeriodicSmoothInitialData	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.zero_isPeriodicSmoothSolutionOn	[propext, Classical.choice, Quot.sound]
NavierStokes3D.KatoLocalWitness.Tpos	[propext, Classical.choice, Quot.sound]
NavierStokes3D.KatoLocalWitness.init	[propext, Classical.choice, Quot.sound]
NavierStokes3D.KatoLocalWitness.pde	[propext, Classical.choice, Quot.sound]
NavierStokes3D.KatoLocalWitness.phys	[propext, Classical.choice, Quot.sound]
NavierStokes3D.KatoLocalWitness.smooth	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_eq_top_iff_forall_lt	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_eq_top_of_bkm	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_eq_top_of_katoRestartTime	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_eq_top_of_unbounded_existenceTimes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_eq_top_of_uniform_restart	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_nonneg	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_pos_of_exists_smooth_solution	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_pos_of_kato_witness	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_zero_eq_top	[propext, Classical.choice, Quot.sound]
NavierStokes3D.VorticityTransportRegularity.pressureC2	[propext, Classical.choice, Quot.sound]
NavierStokes3D.VorticityTransportRegularity.spaceC1	[propext, Classical.choice, Quot.sound]
NavierStokes3D.VorticityTransportRegularity.spaceC2	[propext, Classical.choice, Quot.sound]
NavierStokes3D.VorticityTransportRegularity.spaceC3	[propext, Classical.choice, Quot.sound]
NavierStokes3D.VorticityTransportRegularity.spacetimeC2	[propext, Classical.choice, Quot.sound]
NavierStokes3D.beale_kato_majda	[propext, Classical.choice, Quot.sound]
NavierStokes3D.beale_kato_majda_scaffold	[propext, Classical.choice, Quot.sound]
NavierStokes3D.beale_kato_majda_scaffold_integrability_unused	[propext, Classical.choice, Quot.sound]
NavierStokes3D.bkm_criterion_without_integrability_not_top	[propext, Classical.choice, Quot.sound]
NavierStokes3D.contDiffAt_directionalCoord	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
NavierStokes3D.convective_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_difference	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_energy_pairing_vanishes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_eq_sum_directional	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_norm_le	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_norm_le_of_CZ	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_add	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_add3	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_apply	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_convective	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_convective_div_free	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_gradient	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_laplacian	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_smul	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_time_deriv	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.cyclicLiePairing_expand	[propext, Classical.choice, Quot.sound]
NavierStokes3D.deriv_euclidean_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.deriv_prod_fst_slice	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_convective_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_coord_of_differentiableAt_field	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_curl	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_curl_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_directionalCoord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_half_norm_sq	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_of_contDiffAt_coords	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_pressureGradient_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_time_deriv_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_convective	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
NavierStokes3D.directionalCoord_curl	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_laplacian	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_pressureGradient	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_smul	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_symm	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_add	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_add_smul	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_const	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_curl	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_of_eq_curl	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_smul	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_smul_field	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_zero_field	[propext, Classical.choice, Quot.sound]
NavierStokes3D.energy_zero_of_gronwall	[propext, Classical.choice, Quot.sound]
NavierStokes3D.energy_zero_of_gronwall_Icc	[propext, Classical.choice, Quot.sound]
NavierStokes3D.eq_of_l2_energy_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.eq_zero_of_isPhysicalNSSolution_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.eq_zero_of_isPhysicalNSSolution_zero_all	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ereal_eq_top_of_bkm	[propext, Classical.choice, Quot.sound]
NavierStokes3D.exists_mem_existenceTimes_gt_of_lt_Tstar	[propext, Classical.choice, Quot.sound]
NavierStokes3D.fderiv_apply_fderiv	[propext, Classical.choice, Quot.sound]
NavierStokes3D.fderiv_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.fderiv_directionalCoord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.fderiv_field_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.fderiv_prod_snd_slice	[propext, Classical.choice, Quot.sound]
NavierStokes3D.gradient_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.inner_convective_eq_inner_grad_half_norm_sq	[propext, Classical.choice, Quot.sound]
NavierStokes3D.inner_convective_le_strain	[propext, Classical.choice, Quot.sound]
NavierStokes3D.inner_convective_product_rule	[propext, Classical.choice, Quot.sound]
NavierStokes3D.inner_self_convective_eq_inner_grad_half_norm_sq	[propext, Classical.choice, Quot.sound]
NavierStokes3D.integration_by_parts_of_vanishing_flux	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
NavierStokes3D.isPhysicalNSSolution_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.katoRestartTime_pos	[propext, Classical.choice, Quot.sound]
NavierStokes3D.katoSobolevIndex_lt	[propext, Classical.choice, Quot.sound]
NavierStokes3D.kato_Hs_bound_on_restart_interval	[propext, Classical.choice, Quot.sound]
NavierStokes3D.kato_ponce_implies_quadratic	[propext, Classical.choice, Quot.sound]
NavierStokes3D.kato_ponce_of_uniform_vorticity	[propext, Classical.choice, Quot.sound]
NavierStokes3D.laplacian_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.laplacian_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.le_Tstar_of_mem_existenceTimes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.le_vorticity_sup_norm	[propext, Classical.choice, Quot.sound]
NavierStokes3D.local_existence	[propext, Classical.choice, Quot.sound]
NavierStokes3D.mem_existenceTimes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.mem_existenceTimes_add_of_kato_restart	[propext, Classical.choice, Quot.sound]
NavierStokes3D.mem_existenceTimes_of_kato_witness	[propext, Classical.choice, Quot.sound]
NavierStokes3D.mem_existenceTimes_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.navier_stokes_eq_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ns_cauchy_eq_of_energy_gronwall	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ns_cauchy_of_energy_gronwall	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ns_cauchy_on_Icc_of_energy_gronwall	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ns_momentum_difference	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ns_pde_shift	[propext, Classical.choice, Quot.sound]
NavierStokes3D.one_add_log_le_self	[propext, Classical.choice, Quot.sound]
NavierStokes3D.parabolic_regularity_from_vorticity_bound	[propext, Classical.choice, Quot.sound]
NavierStokes3D.pressureGradient_const	[propext, Classical.choice, Quot.sound]
NavierStokes3D.pressureGradient_sub	[propext, Classical.choice, Quot.sound]
NavierStokes3D.pressure_energy_pairing_vanishes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.reciprocal_of_quadratic_growth	[propext, Classical.choice, Quot.sound]
NavierStokes3D.restart_of_kato_quantitative	[propext, Classical.choice, Quot.sound]
NavierStokes3D.smoothness_below_Tstar_of_uniqueness	[propext, Classical.choice, Quot.sound]
NavierStokes3D.smoothness_of_Tstar_top	[propext, Classical.choice, Quot.sound]
NavierStokes3D.smoothness_on_restart_interval	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
NavierStokes3D.smul_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.strain_sup_le_of_CZ	[propext, Classical.choice, Quot.sound]
NavierStokes3D.stretching_enstrophy_di_at_spatial_max	[propext, Classical.choice, Quot.sound]
NavierStokes3D.stretching_inner_le	[propext, Classical.choice, Quot.sound]
NavierStokes3D.stretching_inner_le_at_max	[propext, Classical.choice, Quot.sound]
NavierStokes3D.stretching_pairing_le_at_spatial_max	[propext, Classical.choice, Quot.sound]
NavierStokes3D.stretching_quartic_le	[propext, Classical.choice, Quot.sound]
NavierStokes3D.sum_univ_fin3	[propext, Classical.choice, Quot.sound]
NavierStokes3D.time_deriv_const	[propext, Classical.choice, Quot.sound]
NavierStokes3D.time_deriv_shift	[propext, Classical.choice, Quot.sound]
NavierStokes3D.time_deriv_sub	[propext, Classical.choice, Quot.sound]
NavierStokes3D.time_space_mixed_partials	[propext, Classical.choice, Quot.sound]
NavierStokes3D.transport_inner_vanishes_of_grad_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.transported_energy_pairing_vanishes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.transported_scalar_along_characteristic	[propext, Classical.choice, Quot.sound]
NavierStokes3D.transported_scalar_maximum_principle	[propext, Classical.choice, Quot.sound]
NavierStokes3D.uniqueness_of_existence_times_of_energy_gronwall	[propext, Classical.choice, Quot.sound]
NavierStokes3D.vorticity_sup_norm_nonneg	[propext, Classical.choice, Quot.sound]
NavierStokes3D.vorticity_transport	[propext, Classical.choice, Quot.sound]
NavierStokes3D.vorticity_transport_at	[propext, Classical.choice, Quot.sound]
NavierStokes3D.vorticity_transport_equation	[propext, Classical.choice, Quot.sound]
NavierStokes3D.vorticity_transport_inner	[propext, Classical.choice, Quot.sound]
Stage6.bochner_leibniz_of_hasDerivAt	[propext, Classical.choice, Quot.sound]
Stage6.hasDerivAt_half_norm_sq	[propext, Classical.choice, Quot.sound]
Stage6.hasDerivAt_inner_self	[propext, Classical.choice, Quot.sound]
Stage6.hasDerivAt_polar	[propext, Classical.choice, Quot.sound]
Stage6.inner_deriv_zero_at_max	[propext, Classical.choice, Quot.sound]
Stage6.polar_pythagoras_of_hasDerivAt	[propext, Classical.choice, Quot.sound]
Stage6.real_iSup_eq_zero_of_not_bddAbove	[propext, Classical.choice, Quot.sound]
Stage6.shearU_hasDerivAt_z	[propext, Classical.choice, Quot.sound]
Stage6.shearU_heat	[propext, Classical.choice, Quot.sound]

Lean declaration	`#print axioms`
`Stage6.shearV_hasDerivAt_z`	`[propext, Classical.choice, Quot.sound]`
`Stage6.shearV_heat`	`[propext, Classical.choice, Quot.sound]`
`Stage6.shear_beltrami`	`[propext, Classical.choice, Quot.sound]`
`Stage6.shear_convective_and_div_zero`	`[propext, Classical.choice, Quot.sound]`
`Stage6.shear_nonconstant`	`[propext, Classical.choice, Quot.sound]`
`Stage6.shear_periodic`	`[propext, Classical.choice, Quot.sound]`
`Stage6.unit_direction_deriv_orthogonal`	`[propext, Classical.choice, Quot.sound]`
`gradient.congr_simp`	`[propext, Classical.choice, Quot.sound]`

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